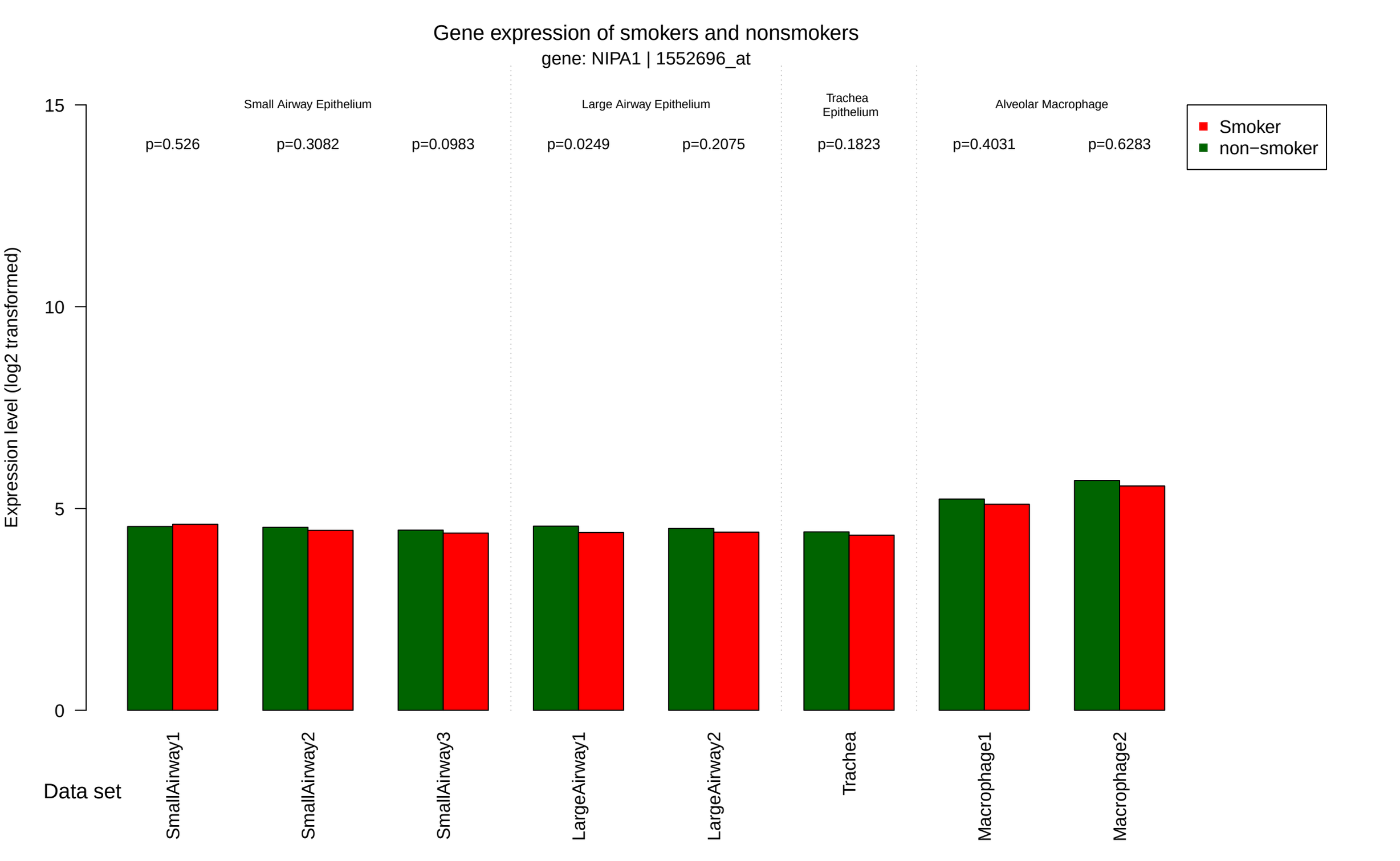
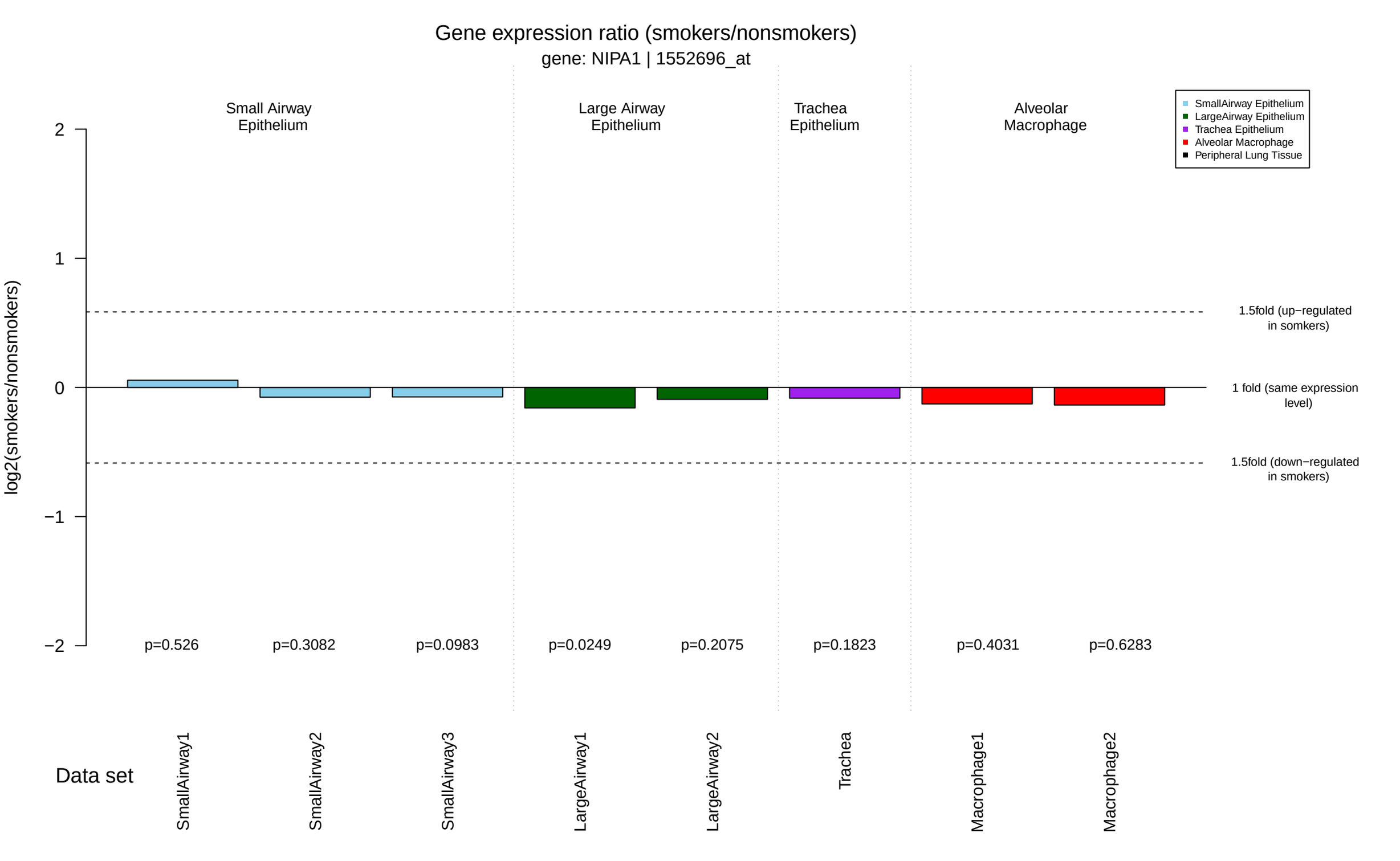
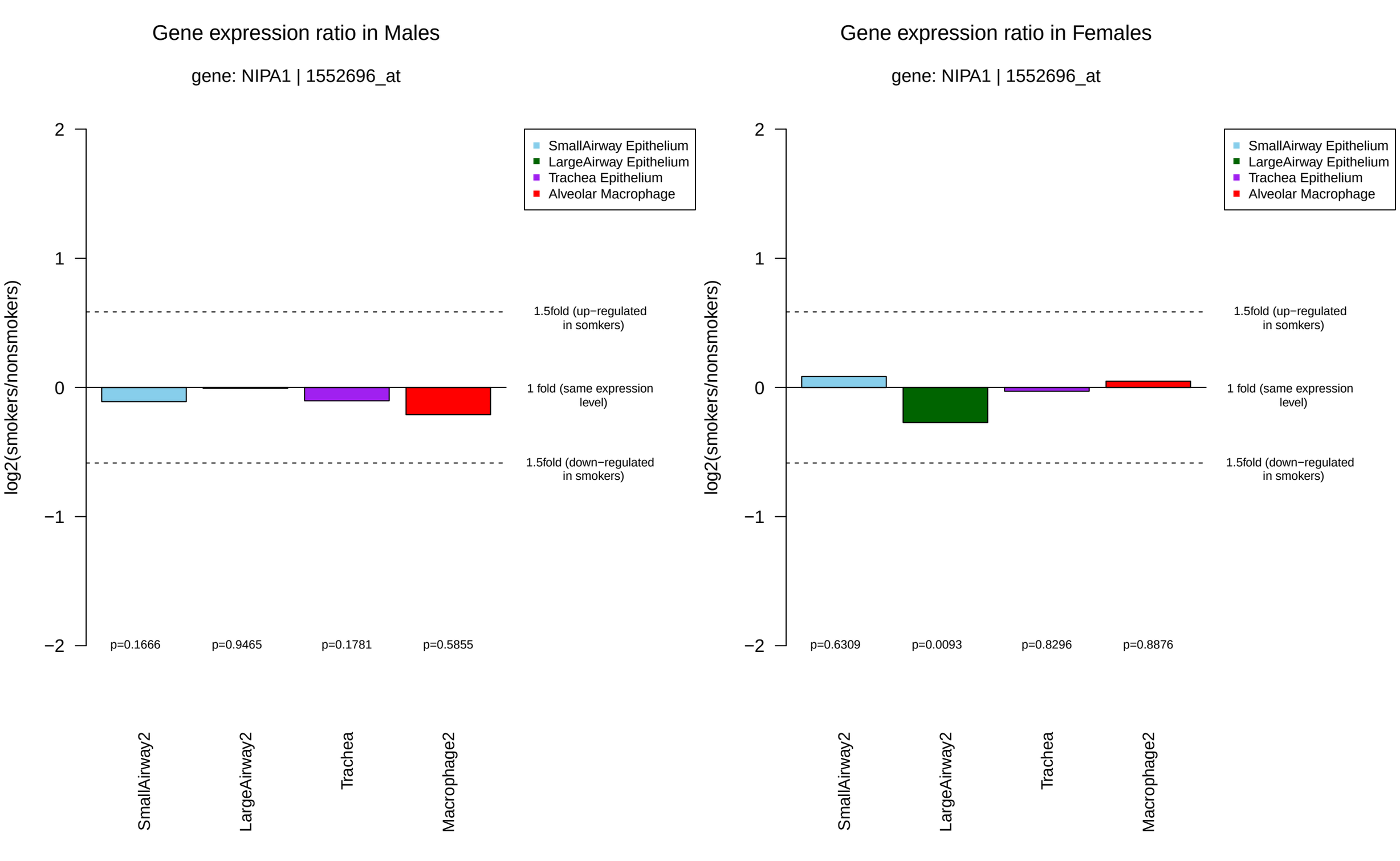
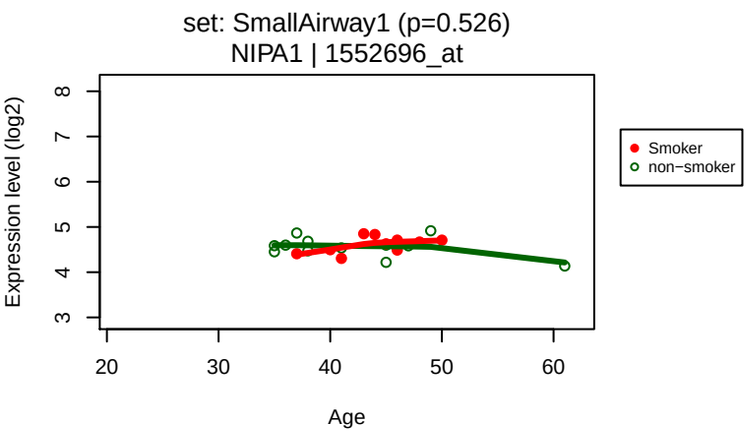
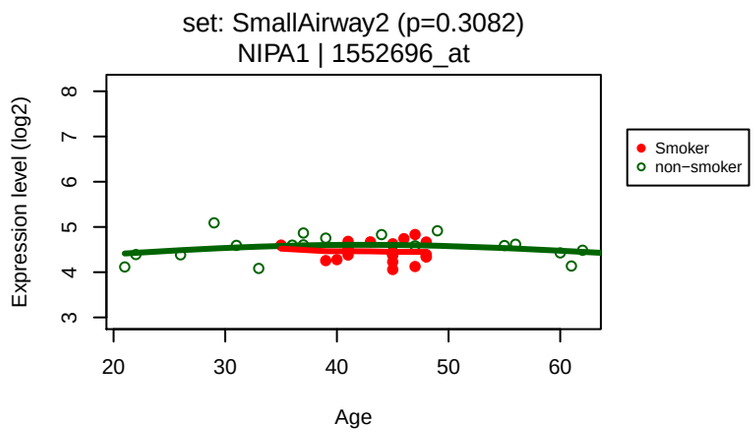
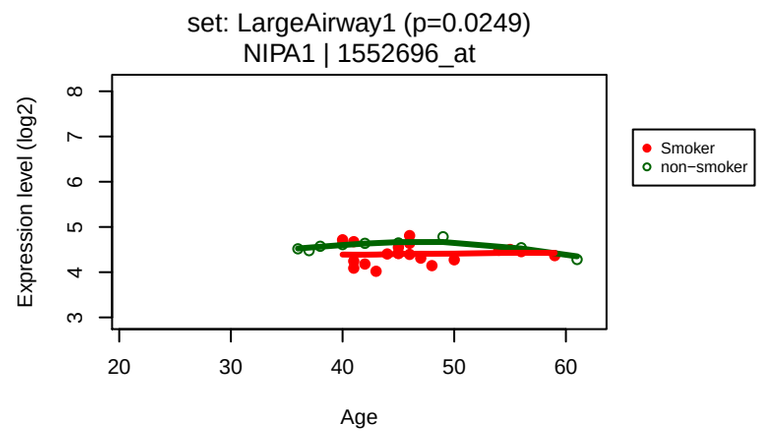

Gene expression of smokers and nonsmokers
gene: NIPA1 | 1552696_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.526
p=0.3082
p=0.0983
p=0.0249
p=0.2075
p=0.1823
p=0.4031
p=0.6283
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

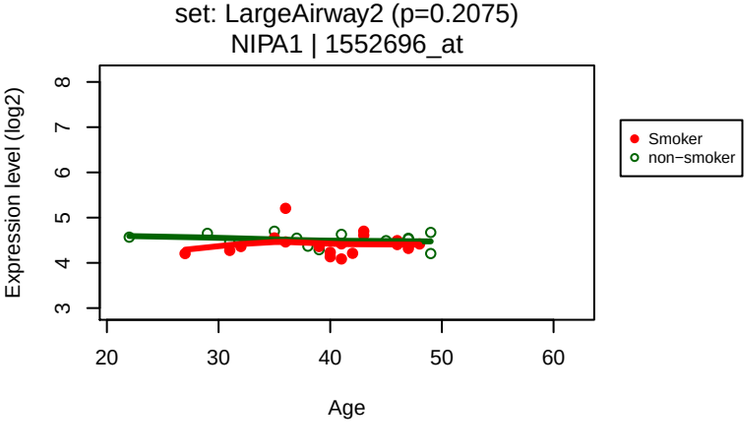

Gene expression ratio (smokers/nonsmokers)
gene: NIPA1 | 1552696_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.526
p=0.3082
p=0.0983
p=0.0249
p=0.2075
p=0.1823
p=0.4031
p=0.6283
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

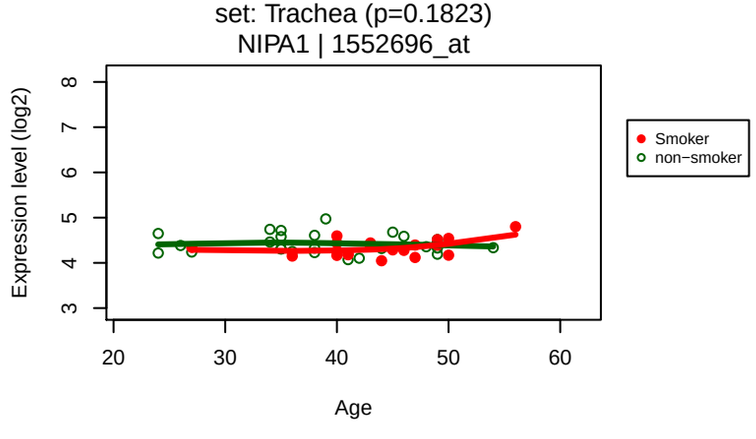

Gene expression ratio in Males
gene: NIPA1 | 1552696_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1666
p=0.9465
p=0.1781
p=0.5855
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NIPA1 | 1552696_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6309
p=0.0093
p=0.8296
p=0.8876
SmallAirway2
LargeAirway2
Trachea
Macrophage2

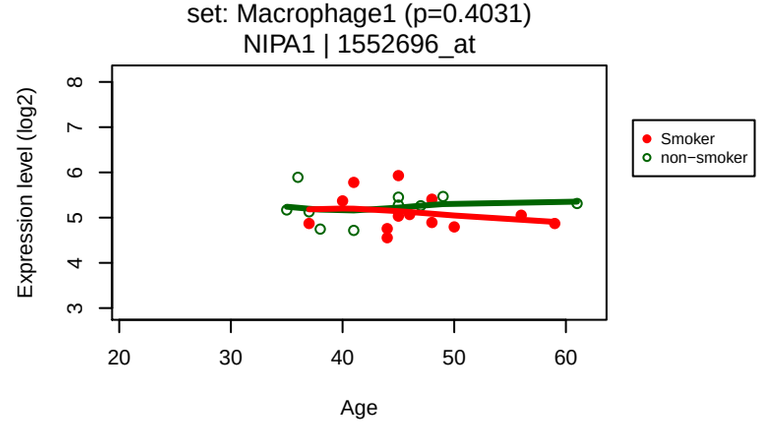

set: SmallAirway1 (p=0.526)
NIPA1 | 1552696_at
Expression level (log2)
Age
Smoker
non−smoker

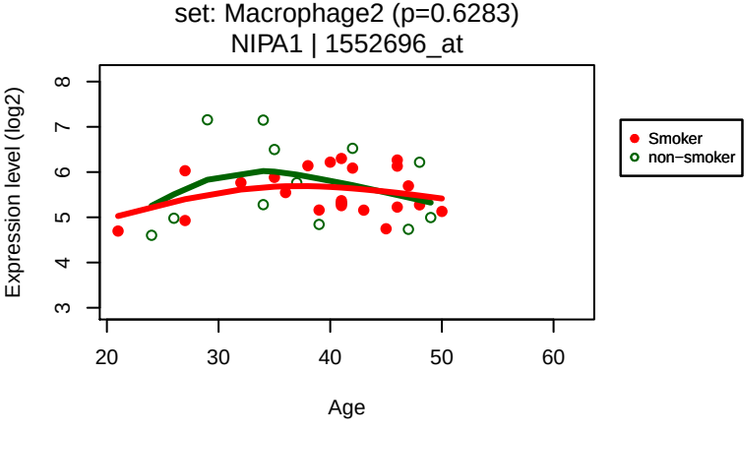

set: SmallAirway2 (p=0.3082)
NIPA1 | 1552696_at
Expression level (log2)
Age
Smoker
non−smoker

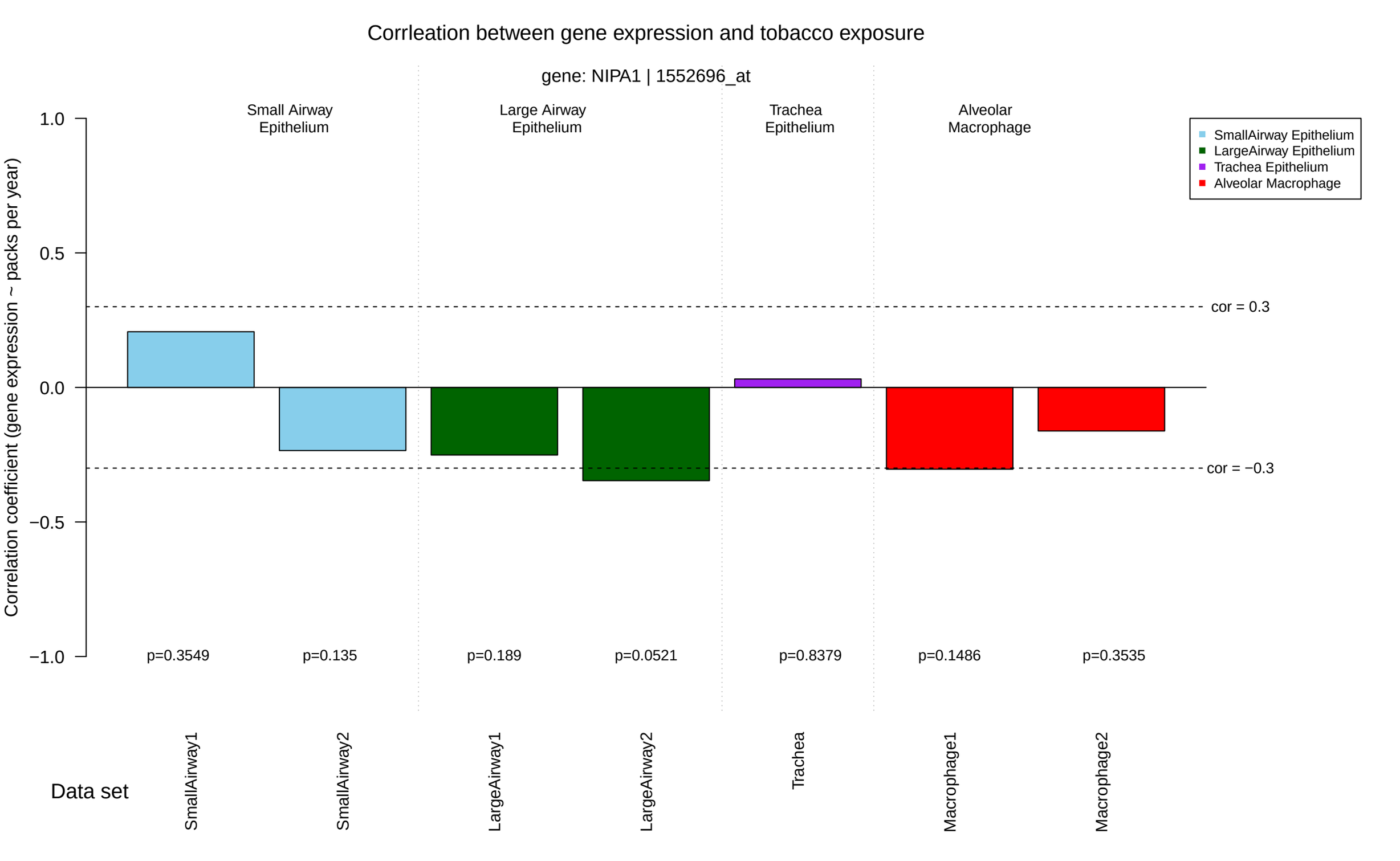

set: LargeAirway1 (p=0.0249)
NIPA1 | 1552696_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2075)
NIPA1 | 1552696_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1823)
NIPA1 | 1552696_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4031)
NIPA1 | 1552696_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6283)
NIPA1 | 1552696_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NIPA1 | 1552696_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3549
p=0.135
p=0.189
p=0.0521
p=0.8379
p=0.1486
p=0.3535
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2